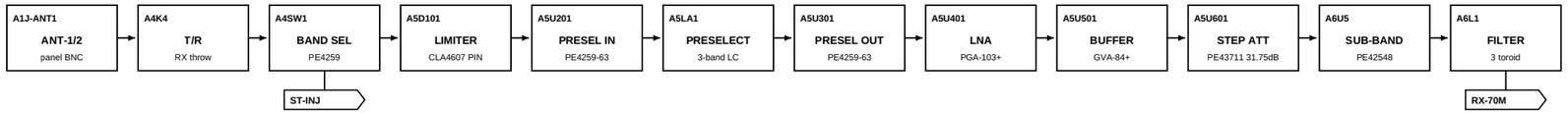
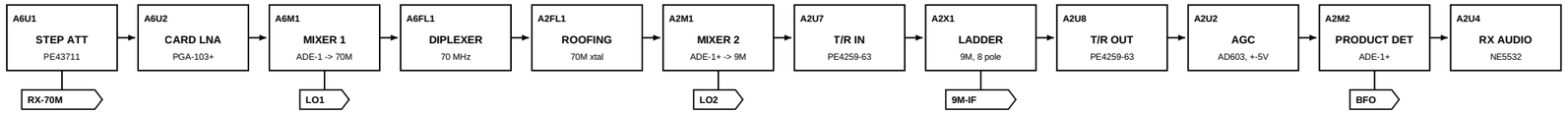
DETAILED BLOCK DIAGRAM. THE WHOLE TRANSCEIVER IN SIX SIGNAL BANDS: RECEIVE, IF AND AUDIO, TRANSMIT, LOCAL OSCILLATOR, CONTROL, POWER.

EACH BLOCK CARRIES ITS ASSEMBLY AND REFERENCE DESIGNATOR. 1.5 TO 470 MHz, SSB/AM/FM, DUAL CONVERSION 70 TO 9 MHz, 20 W PEP.

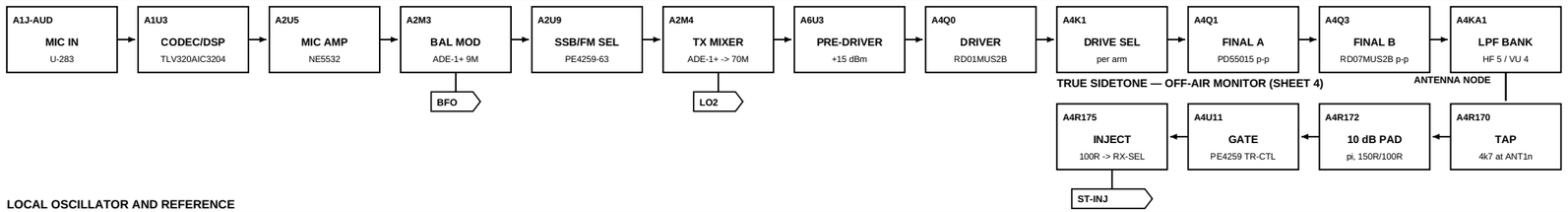
RECEIVE — ANTENNA TO FIRST MIXER



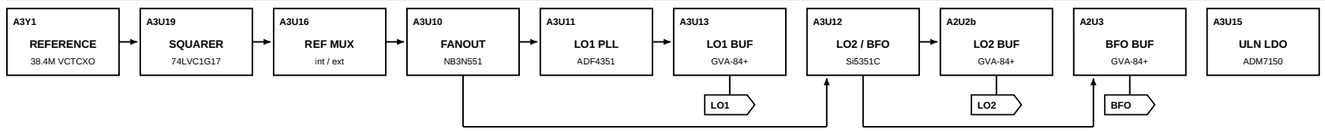
RECEIVE — 70 MHz AND 9 MHz IF, DETECTORS, AUDIO



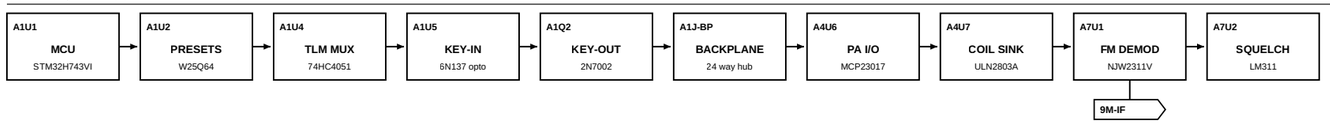
TRANSMIT — MICROPHONE TO ANTENNA



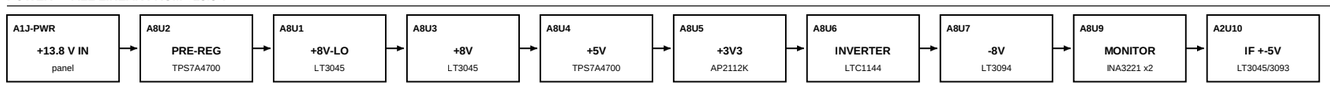
LOCAL OSCILLATOR AND REFERENCE



CONTROL, FM DAUGHTER AND KEYING



POWER — ALL LINEAR FROM +13.8 V



ASSEMBLY SCHEDULE — PART COUNTS READ FROM THE MODEL AT RENDER TIME

ASSY	BOARD	DESCRIPTION	PARTS
A1	MB-FP-01	Front Panel / MCU / Backplane	82
A10	MB-DGT-01	REV 2 Digital Bay	57
A2	MB-IF-01	IF/AF strip Rev D	126
A3	MB-LO-01	LO/Synthesizer	69
A4	MB-PA-01	Power Amplifier	264

ASSY	BOARD	DESCRIPTION	PARTS
A5	MB-RX-01	Wideband RX front end	91
A6	MB-BC-01	Band-Card (one design x4)	110
A7	MB-FM-01	FM Daughterboard Rev C	49
A8	MB-PWR-01	Power	56
A9	MB-RC-J1	J1 entry protection and fan-out	26

REVISIONS		
A	INITIAL RELEASE. SIX-BAND BLOCK DIAGRAM, BLOCKS VERIFIED AGAINST THE MODEL.	2026-11-12

NOTES

1. SIGNAL FLOW IS LEFT TO RIGHT IN EACH BAND. A SOLID LINE IS A SIGNAL PATH; A DASHED LINE IS AN OSCILLATOR INJECTION OR A TAP.
2. EVERY BLOCK NAMES ITS ASSEMBLY AND REFERENCE DESIGNATOR AND IS CHECKED AGAINST BOARDS_COMPONENTS.PY WHEN THIS SHEET IS BUILT. A PART THAT IS DELETED OR RENAMED FAILS THE BUILD AND IS NAMED. THE BLOCKS ARE VERIFIED; THE TOPOLOGY BETWEEN THEM IS A READING OF THE DESIGN, NOT A DERIVATION FROM THE NETLIST.
3. RECEIVE: DUAL CONVERSION, 70 MHZ THEN 9 MHZ. THE NOISE FIGURE IS SET AT A5U401, THE PGA-103+ LNA, AHEAD OF EVERY LOSS THAT FOLLOWS IT. SEE THE RECEIVE CASCADE SHEET FOR LEVELS.
4. THE RECEIVE PATH CROSSES FROM BAND 1 TO BAND 2 ON A6, THE BAND-CARD. IT IS ONE BOARD AND NOT A CABLE; THE WRAP IS A DRAWING CONVENTION ONLY.
5. TRANSMIT SHARES THE 9 MHZ CRYSTAL LADDER WITH RECEIVE THROUGH A2U7 AND A2U8, A PAIR OF PE4259-63 TRANSFER SWITCHES. SSB AND FM ARE SELECTED AT A2U9.
6. THE BAND-CARD A6 IS ONE DESIGN BUILT FOUR TIMES, ONE PER BAND. ONLY ONE IS DRAWN. A6U4 CARRIES THE CARD IDENTITY SO THE MCU CAN TELL WHICH CARD IS IN WHICH SLOT.
7. *A4* CARRIES TWO PUSH-PULL FINALS ON SEPARATE ARMS, EACH WITH ITS OWN LOW-PASS FILTER BANK: FIVE SECTIONS ON THE HF ARM AND FOUR ON THE V/UHF ARM, SWITCHED BY THE RELAY PAIRS KA1..KB4.
8. POWER IS ALL LINEAR FROM +13.8 V. THE OSCILLATOR RAIL IS PRE-REGULATED AT A8U2 AND POST-REGULATED AT A8U1 SO THE SYNTHESISER SEES TWO STAGES OF REJECTION. THE AD603 NEEDS A NEGATIVE RAIL AND GETS IT LOCALLY AT A2, NOT FROM A8.
9. TRUE SIDETONE: A4R170 TAPS THE ANTENNA NODE AFTER THE T/R SWITCH, A4U11 GATES IT ON TR-CTL, AND A4R175 INJECTS IT INTO THE RECEIVE PATH. THE OPERATOR HEARS THE TRANSMISSION ITSELF, DEMODULATED BY THE RADIO'S OWN RECEIVER — PROOF THAT THERE IS MODULATION ON THE OUTGOING CARRIER AND THAT THE RECEIVER WORKS. SHEET 4.
10. ANT-2 AND THE SECOND T/R THROW ARE NOT DRAWN SEPARATELY. THE V/UHF ARM IS IDENTICAL TO THE HF ARM FROM THE T/R RELAY TO THE FILTER BANK.
11. THIS DESIGN IS NOT RELEASED FOR FABRICATION. 312 OF 2396 NET-BEARING PINS HAVE NO PAD ON THEIR BOARD, FOUR PIN ASSIGNMENTS DISAGREE BETWEEN BOARDS, AND THE PRESELECTOR SWITCH IS SPECIFIED AS A PART WITH TOO FEW THROWS. SEE THE OPEN-ITEM REGISTER.

DRAWN 2026-11-12 ANTHONY PELFANIO - K4DIA	CHECKED 2026-11-12 ANTHONY PELFANIO - K4DIA	APPROVED

TITLE — PERMAN-FMR AN/PRC-736

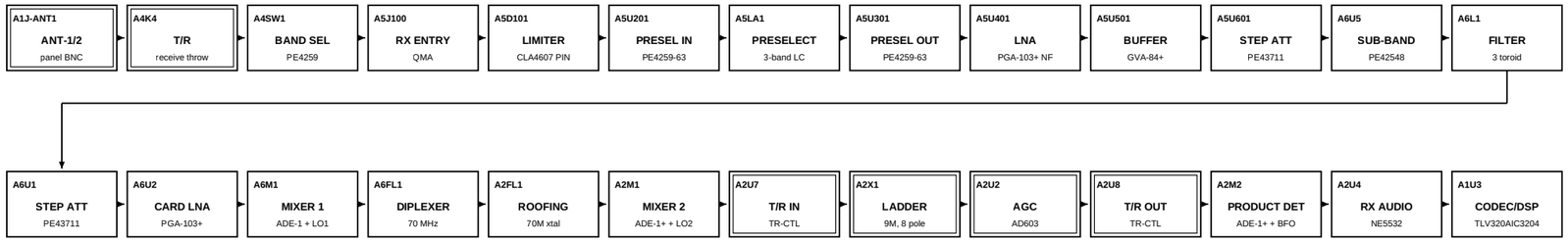
DETAILED BLOCK DIAGRAM

SIZE	DWG NO	REV	SCALE	SHEET
B	**K4DIA-PRC736-001**	**A**	NONE	1 OF 4

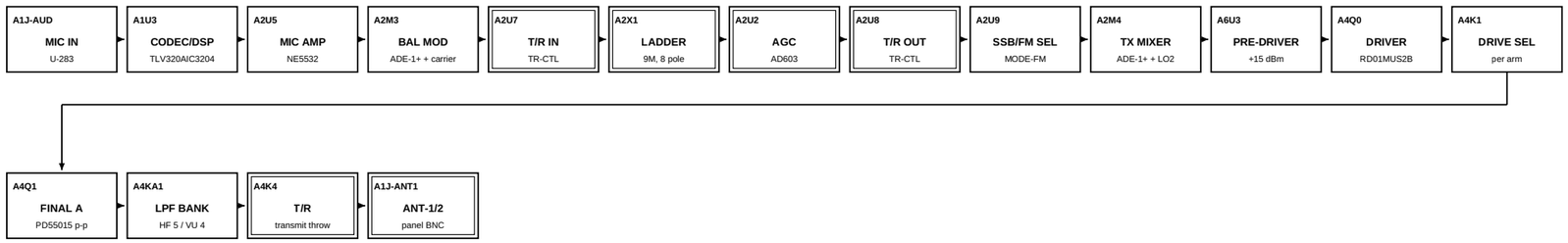
SIGNAL PATH, TRANSMIT AND RECEIVE SEPARATED. EACH CHAIN IS COMPLETE END TO END; STAGES THE TWO PATHS SHARE ARE REPEATED IN BOTH.

A BLOCK WITH A SECOND RULE INSIDE ITS BORDER IS SHARED HARDWARE — SEE SHEET 3 FOR THE SWITCHING THAT HANDS IT BETWEEN THE TWO PATHS.

RECEIVE — ANTENNA TO AUDIO, 26 STAGES



TRANSMIT — MICROPHONE TO ANTENNA, 17 STAGES



REVISIONS		
A	INITIAL RELEASE. TX AND RX DRAWN AS INDEPENDENT CHAINS.	2026-11-12

NOTES

1. EACH CHAIN IS DRAWN COMPLETE. THE SIX STAGES WITH A DOUBLE BORDER ARE THE SAME PHYSICAL HARDWARE IN BOTH AND ARE NOT DUPLICATED IN THE RADIO.
2. RECEIVE IS DUAL CONVERSION: 70 MHZ AT A6M1, 9 MHZ AT A2M1. THE NOISE FIGURE IS SET AT A5U401 AHEAD OF EVERY LOSS THAT FOLLOWS IT.
3. TRANSMIT RUNS THROUGH THE CRYSTAL LADDER AND THE AD603. A2U7 AND A2U8 ARE A PAIR OF PE4259-63 TRANSFER SWITCHES ON TR-CTL AT OPPOSITE ENDS OF THE 9 MHZ SEGMENT, SO THE LADDER AND THE AGC ARE IN BOTH PATHS. ON TRANSMIT THE AD603 IS THE LEVEL CONTROL, NOT A RECEIVE AGC.
4. RECEIVE DOES NOT PASS THE LOW-PASS FILTER BANK. IT LEAVES THE T/R RELAY AT A4K4 FOR THE BAND SELECT AND THE WIDEBAND FRONT END; ONLY TRANSMIT GOES THROUGH A4KA1..KB4.
5. SSB AND FM PART COMPANY AT A2U9, ON MODE-FM. THE FM DAUGHTERBOARD A7 TAKES ITS OWN TAP OFF THE 9 MHZ IF AND RETURNS AUDIO; IT IS NOT IN EITHER CHAIN ABOVE.
6. A THIRD PATH CROSSES BETWEEN THE TWO AND IS NOT DRAWN HERE: THE TRUE SIDETONE, WHICH TAPS THE ANTENNA NODE ON TRANSMIT AND INJECTS INTO THE RECEIVE PATH SO THE RECEIVER DEMODULATES THE RADIO'S OWN TRANSMISSION. SHEET 4.
7. ONLY ONE OF THE FOUR BAND-CARDS IS DRAWN. A6 IS ONE DESIGN BUILT FOUR TIMES, ONE PER BAND.
8. ANT-2 AND THE SECOND T/R THROW ARE NOT DRAWN. THE V/UHF ARM IS IDENTICAL FROM THE T/R RELAY TO THE FILTER BANK.
9. THIS DESIGN IS NOT RELEASED FOR FABRICATION. SEE THE OPEN-ITEM REGISTER.

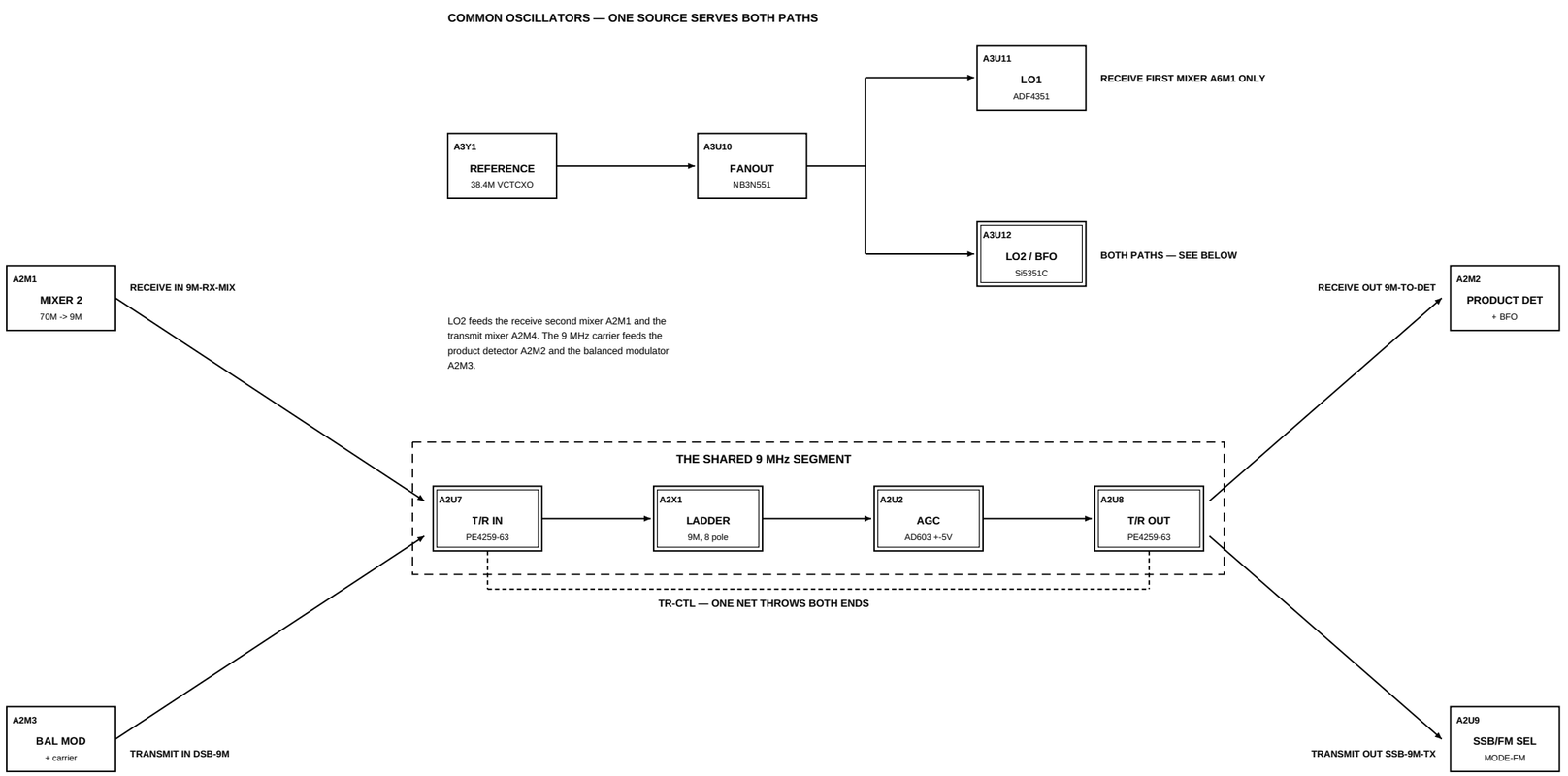
DRAWN	CHECKED	APPROVED
ANTHONY PELFANIO - K4DIA 2026-11-12	ANTHONY PELFANIO - K4DIA 2026-11-12	

TITLE: **SIGNAL PATH, TRANSMIT AND RECEIVE SEPARATED** — PERMAN-FMR AN/PRC-736

SIZE	DWG NO	REV	SCALE	SHEET
B	K4DIA-PRC736-001	A	NONE	2 OF 4

SIGNAL PATH, COMMON HARDWARE. WHAT TRANSMIT AND RECEIVE SHARE, AND THE SWITCH THAT HANDS IT FROM ONE TO THE OTHER.

RECEIVE ENTERS AND LEAVES ABOVE THE CENTRE LINE, TRANSMIT BELOW IT. CONTROL NETS ARE NAMED ON EACH SWITCH.



REVISIONS		
A	INITIAL RELEASE. SHARED STAGES AND THE SWITCHING THAT HANDS THEM OVER.	2026-11-12

NOTES

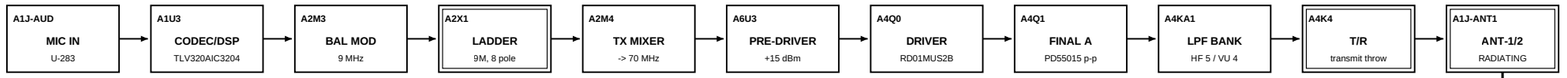
1. THE TWO TRANSFER SWITCHES SIT AT OPPOSITE ENDS OF THE 9 MHZ SEGMENT, SO EVERYTHING BETWEEN THEM IS SHARED: THE EIGHT-POLE CRYSTAL LADDER AND THE AD603. TRANSMIT PASSES THROUGH BOTH.
2. ONE NET, TR-CTL, THROWS A2U7, A2U8 AND THE ANTENNA RELAY A4K4 TOGETHER. THERE IS NO STATE IN WHICH ONE END IS TRANSMITTING AND THE OTHER RECEIVING.
3. ON RECEIVE THE AD603 IS THE AGC. ON TRANSMIT THE SAME PART IS THE LEVEL CONTROL. ITS GAIN LAW AND ITS CONTROL SOURCE DIFFER BETWEEN THE TWO AND ARE NOT INTERCHANGEABLE; SEE THE RECEIVE CASCADE SHEET FOR THE RECEIVE CASE.
4. LO2 AND THE 9 MHZ CARRIER BOTH COME FROM A3U12, ONE SI5351C. A SINGLE PART FAILING TAKES OUT BOTH PATHS AT ONCE, WHICH IS THE COST OF THE SAVING.
5. LO1 IS NOT SHARED. IT SERVES THE FIRST MIXER A6M1 ON RECEIVE ONLY; TRANSMIT REJOINS THE CHAIN AT 70 MHZ AFTER A2M4.
6. THE ANTENNA, ITS RELAY AND THE WIDEBAND FRONT END ENTRY ARE COMMON. THE LOW-PASS FILTER BANK IS NOT: RECEIVE LEAVES A4K4 FOR THE BAND SELECT AND NEVER SEES IT.
7. THE SIDETONE IS THE OTHER TRANSMIT-TO-RECEIVE CROSSING. A4U11 GATES A PADDED SAMPLE OF THE ANTENNA NODE INTO THE RECEIVE PATH ON TR-CTL, THE SAME NET THAT THROWS THESE SWITCHES, SO THE RECEIVER DEMODULATES THE TRANSMISSION WHILE IT IS BEING SENT. SHEET 4.
8. A BLOCK WITH A SECOND RULE INSIDE ITS BORDER IS SHARED HARDWARE, THE SAME CONVENTION AS SHEET 2.
9. THIS DESIGN IS NOT RELEASED FOR FABRICATION. SEE THE OPEN-ITEM REGISTER.

DRAWN 2026-11-12 ANTHONY PELFANIO - K4DIA	CHECKED 2026-11-12 ANTHONY PELFANIO - K4DIA	APPROVED	
TITLE **SIGNAL PATH, COMMON HARDWARE**			PERMAN-FMR AN/PRC-736
SIZE **B**	DWG NO **K4DIA-PRC736-001**	REV **A**	SCALE NONE	SHEET 3 OF 4

TRUE SIDETONE. A SAMPLE OF THE RF ACTUALLY LEAVING THE ANTENNA IS FED BACK INTO THE RECEIVER AND DEMODULATED. WHAT THE OPERATOR HEARS IS THE TRANSMISSION ITSELF.

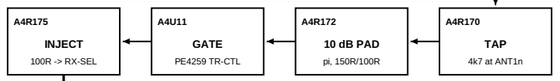
NOT AN AUDIO OSCILLATOR. THE TONE PROVES THERE IS MODULATION ON THE OUTGOING CARRIER AND THAT THE RECEIVER IS WORKING — TWO FAULTS CAUGHT BY ONE LISTEN, WHILE TRANSMITTING.

1 TRANSMIT — MICROPHONE TO THE RADIATING ANTENNA



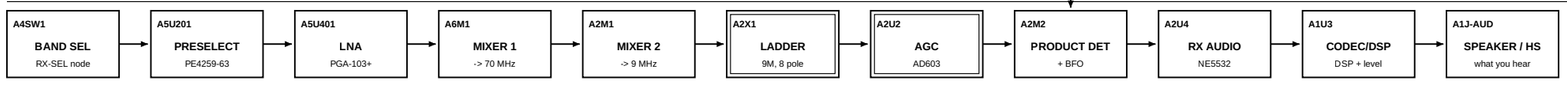
ANTENNA NODE — AFTER THE T/R SWITCH, SO IT IS THE REAL OUTGOING SIGNAL

2 THE SAMPLE — TAPPED AT THE ANTENNA NODE, PADDED, GATED, INJECTED



INJECTED INTO THE RECEIVE PATH AT RX-SEL

3 RECEIVE — THE RADIO DEMODULATES ITS OWN TRANSMISSION



WHAT THE OPERATOR HEARS, AND WHAT IT PROVES

AUDIO PRESENT AND CLEAN — the modulator, the transmit mixer chain, the final and the filter bank are all passing modulation, AND the whole receive chain is working.

CARRIER BUT NO AUDIO — the transmitter is radiating and the modulation is not reaching it. Look at the microphone, the codec and the balanced modulator.

NOTHING AT ALL — either nothing is being radiated or the receiver is dead. The two are told apart by the forward power reading on the same sheet as the ALC.

DISTORTED — overdrive, flat-topping or an ALC fault, heard as it is actually leaving the antenna rather than as it leaves the modulator.

REVISIONS		
A	INITIAL RELEASE. OFF-AIR MONITOR LOOP, ANTENNA NODE TO EARPIECE.	2026-11-12

NOTES

1. THE TAP IS AT THE ANTENNA NODE, AFTER THE T/R SWITCH AND AFTER THE LOW-PASS FILTER BANK, SO THE SAMPLE IS THE SIGNAL THAT IS ACTUALLY GOING TO THE ANTENNA — NOT A SAMPLE OF THE DRIVER, AND NOT A LOCALLY GENERATED TONE.
2. A4R170 TAPS THE HF ARM AT ANT1N AND A4R171 THE V/UHF ARM AT ANT2N. ONLY ONE ARM TRANSMITS AT A TIME; THE IDLE ARM'S TAP LIGHTLY LOADS STONE-T AND DOES NOTHING ELSE.
3. 4K7 IN 1206. AT 9 W OUT THE TAP DISSIPATES ABOUT 93 MW, WHICH IS WHY IT IS A 1206 AND NOT AN 0402.
4. A4U11, A PE4259, GATES THE SAMPLE AND IS OFF IN RECEIVE. IT IS NOT OPTIONAL: WITHOUT IT THIS TAP IS AN ANTENNA-TO-RECEIVER PATH AROUND THE PRESELECTOR, WHICH WOULD UNDO THE FRONT-END SELECTIVITY THE WHOLE BAND-CARD EXISTS TO PROVIDE. IT IS THROWN BY TR-CTL, THE SAME NET AS THE T/R RELAY.
5. A4R175 INJECTS 100R ACROSS A 50 OHM LINE, WHICH COSTS ABOUT 0.4 DB OF RECEIVE INSERTION LOSS. THAT IS THE STANDING PRICE OF THE MONITOR AND IT IS PAID ON EVERY RECEIVE, TRANSMITTING OR NOT.
6. THE RECEIVER IS IN ITS NORMAL CONFIGURATION THROUGHOUT. NOTHING IS SWITCHED FOR MONITORING EXCEPT THE GATE, SO WHAT IS HEARD IS THE RECEIVER AS IT WILL BEHAVE ON THE NEXT INCOMING SIGNAL.
7. THE AUDIO RETURNS THROUGH THE CODEC AT A1U3, SO THE DSP SEES THE MONITORED TRANSMISSION AND THE LEVEL CAN BE SET INDEPENDENTLY OF RECEIVE VOLUME.
8. MOST MODERN RADIOS DO NOT DO THIS. A CONVENTIONAL SIDETONE IS AN OSCILLATOR IN THE AUDIO CHAIN: IT CONFIRMS THE KEY CLOSED AND NOTHING ELSE. IT CANNOT TELL THE OPERATOR THAT THE POWER AMPLIFIER IS RADIATING, THAT THE MODULATION IS ON THE CARRIER, OR THAT THE RECEIVER WILL COME BACK WHEN THE RELAY DROPS.
9. THIS DESIGN IS NOT RELEASED FOR FABRICATION. SEE THE OPEN-ITEM REGISTER.

DRAWN 2026-11-12	CHECKED 2026-11-12	APPROVED
ANTHONY PELFANIO - K4DIA	ANTHONY PELFANIO - K4DIA	

TITLE — PERMAN-FMR AN/PRC-736

TRUE SIDETONE, OFF-AIR MONITOR LOOP

SIZE	DWG NO	REV	SCALE	SHEET
B	K4DIA-PRC736-001	A	NONE	4 OF 4